PROTOCOL AND ETHICS APPROVAL FOR HEALTH FACILITY ADAPTATION INTERVENTIONS

Deliverable 5.4

Funded by the European Union

UKRI Innovate UK

Protocol and ethics approval for health facility adaptation interventions

HIGH HORIZONS – D5.4

[GRANT NO. 101057843]

Deliverable Information	
Title	Protocol and ethics approval for health facility adaptation interventions: Evaluation of HIGH Horizons Climate Adaptation Intervention in South Africa and Zimbabwe
Deliverable number	D5.4
WP number	5
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Lead beneficiary	LSHTM
Type	Report
Dissemination Level	Public
Due date	31 December 2024 New due date: 28 February 2025

History of Changes	
Version 0.1	Protocol developed and submitted by LSHTM; 31 October 2024**
Version 0.2	Protocol completed with adaptation interventions selected in South Africa and Zimbabwe (31 January 2025)
Version 0.3	Internal review; 27 February 2025
Version 1.0	Final document; 28 February 2025 ** The protocol has been prepared for submission to the ethics committee at LSHTM and adapted to the local requirements for submission to the respective ethics committees in South Africa and Zimbabwe.

This document is issued within the frame and for the purpose of the HIGH Horizons project. This project has received funding from the European Union's Horizon Framework Programme under Grant Agreement number 101057843. Project partner LSHTM is funded by UKRI Innovate UK reference number 10038478. The opinions expressed and arguments employed herein do not necessarily reflect the official views of the European Commission or UKRI. The dissemination of this document reflects only the author's view, and the European Commission and UKRI are not responsible for any use that may be made of the information it contains. This deliverable is subject to final acceptance by the European Commission.
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Acronyms

Acronym	Definition
AKDN	Aga Khan Development Network
ANC	Antenatal care
aOR	Adjusted odds ratio
ARIMA	Autoregressive integrated moving average
CCA	Cost consequence analysis
CeSHHAR	Centre for Sexual Health and HIV AIDS Research
CHC	Community health centre
CI	Confidence interval
DTU	Technical University of Denmark
EHT	Heat champions (South Africa)
EU	European Union
EWS	Early Warning System
FGD	Focus Group Discussion
GDPR	General Data Protection Regulation
GHCC	Global Health Costing Consortium
HCW	Healthcare worker
HREC	Health Research Ethics Committee
IDIs	In-depth Interviews
IEC	Information, Education and Communication
LMIC	Low- and middle-income country
LSHTM	London School of Hygiene and Tropical Medicine (United Kingdom)
MNH	Maternal and Neonatal (or Newborn) Health
MRC	Medical Research Council (United Kingdom)
MRCZ	Medical Research Council of Zimbabwe
NIHR	National Institute for Health Research
OHS	Occupational Health Standard
PNC	Postnatal care
POPIA	Protection of Personal Information Act (South Africa)
QoC	Quality of care
SA	South Africa
SOP	Standard Operating Procedures
STAMP	Suggested Time-motion Procedures
UNFPA	United Nations Population Fund
UNICEF	United Nations agency for Children
WOMBAT	Work Observation Method by Activity Timing tool
WHO	World Health Organisation
WP	Work package

1 Introduction

Climate change is increasingly impacting maternal and newborn health through hazards like high temperatures, droughts, and flooding. The HIGH Horizons project focuses on the direct and indirect effects of elevated temperatures on the health and wellbeing of women, newborns, young children and on healthcare workers (HCW). Research by Conway et al. (2024) and Lakhoo et al. (2024) indicate that higher temperatures increase risks for adverse pregnancy outcomes, such as pre-eclampsia, gestational diabetes, preterm births, stillbirths and mental health issues. The quality of care provided by healthcare workers, who themselves suffer from heat strain, is likely to significantly influence these outcomes.

Healthcare workers in hot environments experience reduced productivity and wellbeing, potentially leading to errors, delays and mistreatment in patient care and suboptimal outcomes. In African settings, where resources are limited, these impacts are likely to be exacerbated. Despite this, research on how heat affects healthcare workers is scarce, particularly in low- and middle-income countries (LMIC), with only three existing studies in 157 identified by Amoadu et. al.'s systematic review (2023).

The HIGH Horizons project aims to develop climate adaptation mitigation interventions in healthcare facilities across South Africa and Zimbabwe. The primary focus is on improving the wellbeing, performance, and quality of care provided by healthcare workers. The project integrates interdisciplinary scientific methodologies with stakeholder input to create cost-effective and locally relevant solutions.

As part of this approach, a mixed-method baseline study was conducted from August 2023 to September 2024, in both hot and cold seasons in South Africa and Zimbabwe. The following ethics committees approved this baseline study: LSHTM (reference 28413), University of the Witwatersrand (Wits HREC reference 230508), and Medical Research Council of Zimbabwe (MRCZ/A/3023). This baseline study

culminated in the co-design of multi-component climate adaptation interventions in South Africa and Zimbabwe. In both countries, the co-design process began with a participatory stakeholder analysis workshop in July 2024 to identify key stakeholders for inclusion in subsequent co-design workshops. In South Africa, the co-design workshops were conducted in July and August 2024 and the intervention design was completed in August 2024, while in Zimbabwe, a series of co-design workshops were conducted between December 2024 and will be completed by February 2025.

In parallel facility-based mitigation interventions were developed using the CARBOMICA tool, a resource allocation tool for carbon mitigation interventions in healthcare facilities (Luchters, Machingura, et al., 2024). The CARBOMICA tool integrates stakeholder insights, procurement data, and epidemiological, behavioural, budgetary, and programmatic data in its model. By leveraging carbon emission hotspots identified through the AKDN carbon tracking tool, validated by World Health Organisation (WHO) and adhering to high-impact carbon reduction principles, CARBOMICA aligns with mitigation interventions recommended by health facility managers and supported by robust impact size data.

Following these activities, an endline study and a process evaluation are proposed to evaluate the effectiveness and the cost-consequences of the multi-component climate adaptation-mitigation interventions in each country. *The purpose of this protocol is to present the methodology for evaluating the HIGH Horizons co-designed climate mitigation and adaptation interventions in South Africa and Zimbabwe.*

2 Research team

This protocol is part of a collaboration between four research teams within the HIGH Horizons project. The main members of these teams with responsibility with various aspects of the proposed evaluation are listed in Table 1 below.

Table 1. Research team for the evaluation of the HIGH Horizons adaptation interventions

Wits RHI	CeSHHAR	LSHTM	DTU
▪ Shobna Sawry: deputy-local PI; intervention and evaluation design & implementation ▪ Gloria Maimela: local PI; intervention and evaluation design ▪ Celeste Madondo: overall programme management, quality control and planning ▪ Rose Lamola: project manager and implementation oversight ▪ Pascalia Munyewende: qualitative study, early warning system oversight and management ▪ Jean Le Roux: data management, thermal monitoring and carbon mitigation oversight ▪ Sibusiso Mkwananzi: quantitative methods, evaluation of interventions, healthcare worker study evaluation oversight and management ▪ Statistician: data analysis	▪ Fortunate Machingura: local PI, intervention and evaluation design and implementation ▪ Stanley Luchters: overall consortium PI, overall study design ▪ Brian Sibanda, Tariro Chinozvina, Concilia Mutasa: qualitative study, process evaluation ▪ Tapiwa Nyakabau: quality oversight ▪ Elizabeth Dangaiso, Bongani Mutimutema, Samson Chikura: quantitative survey data collection, time-motion, process evaluation ▪ Progress Chaworeka: cost consequences ▪ Thabani Muronzie: thermal conditions ▪ Jetina Tsvaki: mitigation intervention evaluation	▪ Veronique Filippi: overall study design ▪ Nasser Fardousi: support with survey & time-motion ▪ Isabelle Lange: support with qualitative studies ▪ Giulia Greco: cost consequences ▪ Olimpia Lamberti: cost consequences ▪ Paul Lokubal: support with statistics ▪ Giorgia Gon: time-motion, process evaluation	▪ Jørn Toftum: thermal conditions

3 Baseline study findings

The baseline study was conducted in 5 facilities at different levels of the healthcare pyramid: an urban community health centre (CHC) and an urban regional hospital in South Africa's Tshwane district; a hospital and two rural facilities (health centre and clinic) in Zimbabwe's Mount Darwin district. A total of 72 HCWs in South Africa and 53 in Zimbabwe participated across methods. The thermal conditions in the five healthcare facilities have been monitored since Autumn 2023 and now span both hot and cool seasons.

The preliminary analysis of data from both countries' sites indicates that heat has a wide range of impacts, most noticeably on heat strain and performance; however, we found mixed and contradictory results for other outcomes such as dehydration and mental health (see Table 2). We provide further details on these findings below.

Table 2. Summary of the preliminary level of evidence of effect of hotter temperature on main outcomes in baseline study

Main outcomes	South Africa				Zimbabwe			
	Qualita-tive	Ques-tion-naire	Time-motion	Other (USG, routine data)	Qualita-tive	Ques-tion-naire	Time-motion	Other (USG, routine data)
Dehydration				Some				Yes
Psych stress	Yes	Some			Yes	No		
Wellbeing	Yes	No			Yes	No		
Anxiety	Yes	No			Yes	No		
Cognition	Yes				Yes			
Heat strain	Yes	Yes			Yes	Yes		
Performance/ QoC	Yes	Yes	Pending	Pending	Yes	Yes	Pending	Pending
Respectful care	Yes	Yes	Pending		Yes	Yes	Pending	

Note: Cells shaded in grey indicates that it is not measured by this method; cells shaded in yellow indicate borderline significance or significance achieved only for some sub-analysis; ethnographic study was only conducted in the hot season in South Africa; for anxiety, stress and wellbeing, qualitative methods were coded as yes, if at least one symptoms and signs present in the tools were observed or reported.

Qualitative methods: In *South Africa*, 15 HCWs participated in in-depth interviews, and 157 hours of observations were conducted (Munyewende et al., forthcoming). Our qualitative findings indicate that HCWs face considerable heat-related challenges. Overcrowding in small working areas, involving both staff and patients, intensified the heat burden during the observation period. Key contributing factors included poor ventilation, inconsistently functioning air conditioning, unmaintained water coolers, and non-operational fans. These conditions exacerbated both the physical and mental toll on HCWs.

Physically, HCWs reported experiencing fatigue, dizziness, lethargy, discomfort, dehydration, and skin problems, all of which impeded their ability to perform tasks. Mentally, heat exposure led to increased irritability, anger, stress, and an overall decline in their cognitive functions and mental wellbeing, which further affected their work performance, interactions with colleagues and patients, and ultimately the quality of care (QoC) provided.

To cope with the heat, HCWs employed various strategies: standing in front of operational fans, using personal hand-held fans, seeking cooler indoor areas, stepping outside for fresh air, and turning off lights. Air conditioning appeared to provide relief but was inconsistently used, with nurses reporting disagreements among HCWs on its usage. Additionally, staff expressed concerns that air conditioning in the maternity unit could be harmful to newborns.

In *Zimbabwe*, 24 HCWs participated in the qualitative study (Machingura et al., 2024). Heat exposure was found to impair HCWs' physical coordination, decision-making, and precision. They reported symptoms such as fatigue, headaches, light-headedness, and reduced focus during deliveries, which compromised their ability to perform tasks like suturing and maintaining accurate records. The negative impact of heat stress was most evident in high-pressure environments, such as delivery wards, where HCWs are responsible for both the mother and the safe delivery of the newborn.

Healthcare workers in antenatal care (ANC), delivery, and postnatal care (PNC) wards have incorporated heat adaptation strategies into their daily routines due to constant exposure to high temperatures and challenging working conditions. Heat is perceived as a normal aspect of the work environment, which diminishes the attention given to addressing its effects on HCW performance and patient care. One midwife noted that concerns about providing essential maternal and newborn health (MNH) care often take precedence over the need for thermal comfort, despite repeated heat exhaustion negatively impacting mental health, performance, and increasing the likelihood of errors.

In Zimbabwe, psychological preparedness appears to be a key adaptation strategy, with HCWs mentoring new staff on how to cope with the heat, as resources like cooling systems are often lacking. Simple measures, such as using wet towels or standing in front of fans (if available), are commonly employed. Despite these coping strategies, the lack of systemic adaptations—such as adequate cooling equipment and scheduled breaks—appeared to exacerbate the negative effects of heat stress on HCWs' wellbeing and the quality of care they provide.

Questionnaire survey: While the questionnaire survey found significant variations for several heat strain indicators, such as being tired, feeling thirsty and feeling uncomfortable, between the hot and cold seasons, it did not find evidence of an impact on wellbeing using the validated tools anxiety (GAD-7) and quality of life (WHO-5 wellbeing index). For general anxiety stress disorder (PSS-4), only a very borderline statistical difference was found between hot and cold seasons in South Africa (p-value: 0.09). The respondents in both countries noted that almost all indicators of quality of care were impacted by heat, ranging from reduced productivity to irritability, and that most were not aware of policy specific to heat in their facilities.

Time-motion: Our time-motion data analysis is ongoing and appears to corroborate findings on the impact of health on quality of care from the other

methods. We evaluated the impact of heat on the quality of care by analysing how heat influenced the proportion of time a midwife spent inside the delivery cubicle relative to the total session time (Reddy, 2024). This indicator serves as a proxy for a positive care experience, a key domain of quality maternal services as defined by the World Health Organisation. We found that higher temperatures were associated with a decrease in the average percentage of time a midwife spent in the room (aOR: 0.976 [95% CI: 0.955-0.998]). In South Africa, this effect translated to a reduction of 0.6 percentage points (95% CI: [0.04-1,1], p-value: 0.035). We are still in the process of conducting a more exhaustive analysis of the time-motion data.

Other methods: We also assessed dehydration levels by measuring mean urine-specific gravity. In *Zimbabwe*, a significant difference in dehydration levels was observed between hot and cold seasons, with 75% of providers dehydrated in the hot season compared to 59% in the cooler season (p-value = 0.065). The median urine-specific gravity was 1.024 in the hot season versus 1.022 in the cool season (p-value = 0.032). In *South Africa*, no significant difference in dehydration levels was found between hot and cold seasons overall. However, providers in South Africa were significantly more dehydrated during periods of higher temperatures in the hot season (Radebe et al., 2024). We have also extracted information from registers and medical records, and this analysis is ongoing.

In conclusion, most of the effects of high temperatures were consistently found to impact HCW performance and quality of care, as well as contribute to heat strain. Nevertheless, based on these findings, we propose continuing with the same mixed-method approach and using the same range of indicators, even though significant differences were not always observed between cold and hot seasons. This is because the evaluation will focus on the intervention's effect during the hot season only.

4 Development of intervention

We developed the HIGH Horizons climate adaptation-mitigation intervention package using not only data from our mixed-method studies in South Africa and Zimbabwe, but also thermal condition modelling data (Toftum et al., 2024), the literature and the CARBOMICA modelling tool. The intervention's co-design approach and evaluation can be mapped out on the six steps in quality intervention development (6SQuID) (Wight et al, 2015) and the six core elements of the MRC/NIHR framework for developing and evaluating interventions (Skivington et al, 2021), as outlined in Table 3. As the research teams in South Africa (Wits RHI) and Zimbabwe (CeSHHAR) are concurrently involved in a Wellcome Trust funded project called HAPI, focusing on heat adaptation at individual, household and community level, the co-designing of activities involved both projects.

Table 3. Steps and elements in the development of the HIGH Horizons interventions in South Africa and Zimbabwe, and how they map out on the 6SQuID and the MRC/NIHR frameworks

6SQuID framework	MRC/NIHR framework: six core elements	Activity	Status
Define and understand the problem and its cause	How does the context influence the problem and its causes?	Mixed method health worker study	Completed in April 2024 in SA, and July 2024 in Zimbabwe
		Monitoring of thermal conditions	Completed for baseline; ongoing for endline
		Confirmation of link between thermal conditions, providers wellbeing, performance and quality of care	Preliminary findings completed in October 2024
Clarify which causal and contextual factors are malleable and have greatest scope for change	What is the underpinning programme theory? What are the key uncertainties?	Modelling of thermal conditions	Completed in October 2024
		Workshop 1 with HIGH Horizons international research team in London; preparing initial conceptual framework	May/June 2024
		Workshop 2 with HAPI and HIGH Horizons research teams in South Africa and Zimbabwe synthetising	July 2024

6SQuID framework	MRC/NIHR framework: six core elements	Activity	Status
	How can diverse stakeholders' perspectives be included in the research?	results from health facilities and communities together	
Identify how to bring change: the change mechanism		Workshop 3 with stakeholders in Zimbabwe and South Africa (Prioritising and selecting package of interventions; preparing theory of change)	August 2024 in South Africa January 2025 in Zimbabwe
Identify how to deliver the change mechanisms		Follow up country workshops and consultations with stakeholders in Zimbabwe and South Africa (Considering feasibility, finalising intervention package for testing and preparing log-frame and theory of change)	August-September 2024 in South Africa January–February 2025 in Zimbabwe
Test and refine on small scale	How does the intervention interact with its context? How can this intervention be refined?	Proposal for evaluation /ethical approval	October-November 2024
		Ongoing refinement during implementation and process evaluation	From December 2024 in South Africa From April 2025 in Zimbabwe
Collect sufficient evidence of effectiveness to justify (further) rigorous evaluation/ implementation	What are the comparative resource and out-come consequences of the intervention?	Full effectiveness and cost-consequences evaluation	From March 2025 From June 2025 in Zimbabwe

In addition, HIGH Horizons is also involved in developing mitigation interventions to reduce carbon emissions in health facilities (Luchters, Machingura, et al., 2024). While these mitigation interventions will be evaluated using a different methodology as part of a separate protocol (Luchters, Muronzie, et al., 2024), this evaluation protocol will also consider the trade-offs and synergetic effects of the climate adaptation and mitigation interventions within the health facilities.

5 Theory of change of the adaptation interventions

This section focusses on the components of the adaptation interventions in South Africa and in Zimbabwe following completion of the co-design processes. Further details about inputs, expected outputs, outcomes, and goals are incorporated in the theory of change which can be found in Appendix 1 for South Africa and Appendix 2 for Zimbabwe. The interventions in both South Africa and Zimbabwe will be similar in terms of the main components. Tables 4 and 5 summarise the interventions components and problems addressed in South Africa and Zimbabwe respectively.

Table 4. Summary of main climate adaptation intervention components which have been selected through the co-design process in South Africa

Type of inter-vention	Intervention component in South Africa	Problems addressed
Human resources/ behaviour	Heat Champion (2 per facilities) Their duties: • Health talks to PPW and HCW • Distribute IEC materials • Aircon controls • Ventilation checks, • Provision of H2O to HCW • Water cooler refilling • Provide HCWs with personal cooling items, such as cooling towels, vests, or fans • Use posters, flyers, and brief educational meetings to emphasise the risks of heat stress and dehydration • Indoor thermal monitoring • Monitoring heat stress symptoms/incidents amongst HCW • Present a tray of cooling device options for personal use (small rechargeable fan, additional water, cooling vest) • Check functioning of cooling devices in facility	• Lack of HCW awareness and education on the impact of heat on their health • Lack of HCW awareness on available adaptation interventions • Lack of access to Heat Health IEC materials • Suboptimal/Incorrect use of air conditioner • Limited access to clean and safe water as HCWs do not take water breaks and do not trust water sources

Type of inter-vention	Intervention component in South Africa	Problems addressed
	• Number of HCWs receiving guidance or interventions from the Heat Champion per month • Encourage HCWs to take water breaks during shifts • Encourage a buddy system where HCWs remind each other to drink water, take breaks, and stay cool • Monitor usage of hydration stations	
Equipment/ behavioural change	Provide water filter cooler Details: • 2-3 water filters per clinic providing cool drinking water for HCW during hot weather in ideal locations	• Lack of access to safe and clean water • Dehydration in HCWs on duty
Building/ thermal conditions	Air conditioners Details: • Solar-powered/low emission/energy efficient air conditioners installed in Stanza CHC (antenatal care waiting room)	• Lack of air conditioners in some areas of Stanza CHC e.g. ANC • Sub-optimal use of air conditioners where available at Stanza CHC • However: Air conditioners powered by electricity and contributing to high gas emissions
Building/ thermal conditions	Reflective roof paint Details: • Identification of ideal locations and acceptable colour for application of reflective roof paint at Laudium CHC, Stanza CHC and Mamelodi Hospital MNH service points • Purchase reflective paint for the roofs of Laudium CHC, Stanza CHC and Mamelodi Hospital	• High indoor temperatures in MNH service points due to non-reflection of heat in uncoated roofs
Building/ thermal conditions	Roof/Ceiling insulation Details: • Install roof insulation at ANC, Labour and Neonates in Stanza CHC and possibly Laudium • Secure quotes, purchase and install insulation in roofs of Stanza and/or Laudium CHC	• High indoor temperatures in MNH areas at Stanza CHC due to uninsulated roofs

Type of inter-vention	Intervention component in South Africa	Problems addressed
Monitoring/ policy	Occupational health standards assessment Details: • Review Occupational Health Standard (OHS) indicators for health facility compliance • Define 2 heat indicators for inclusion as part of OHS facility compliance • Assess the current compliance levels at Stanza and Laudium CHCs and Mamelodi Hospital with established OHS indicators	• Heat indicators are not included/incorporated into routine reporting at facilities
Nature / thermal conditions	Planting trees Details: • At least 50 trees planted in Stanza Bopape CHC and Mamelodi Hospital in high traffic areas accessible to MNH HCWs (e.g. in close proximity to wards)	• Lack of trees and vegetation increase heat islands in facilities • Lack of tree shading for HCWs

Table 5. Summary of main climate adaptation intervention components which have been selected through the co-design process in Zimbabwe

Type of inter-vention	Intervention component in Zimbabwe	Problems addressed
Behaviour/ Individual	1. Phone based Heat EWS 2. Signage reminders to maintain hydration 3. Buddy system 4. Adjusting work schedules 5. Training on heat related conditions 6. Personal cooling kits	1. Lack of heat management 2. Lack of knowledge of impending heat waves/ high temperatures 3. Dehydration in HCW on duty 4. Dehydration in HCW on duty 5. Dehydration and lack of heat management 6. Lack of knowledge of heat-related conditions, including heat exhaustion and heatstroke. Dehydration and lack of heat management
Behaviour/ Facility-level	1. ANC Heat-Health talks 2. Heat management training 3. Indoor thermal monitoring 4. Waste segregation 5. Heat champions - EHTs	1. Lack of knowledge of heat-related conditions, including heat exhaustion and heatstroke 2. idem 3. Dehydration 4. Waste mismanagement 5. Lack of heat management

Type of inter-vention	Intervention component in Zimbabwe	Problems addressed
Behaviour/ Policy	Develop facility guidance to support heat-health training, heat-adaptive work schedules, rests, lightweight clothing and heatwave emergencies	• Lack of knowledge of heat-related conditions, including heat exhaustion and heatstroke • Dehydration • Lack of heat management
Built Environ-ment	1. Install solar-powered, energy-efficient air conditioners in ANC waiting rooms. 2. Construct shaded rest areas, such as thatched outdoor spaces, for HCWs. 3. Enhance ventilation in MNH workstations using louvres and cross-ventilation systems. 4. Apply reflective paint on roofs 5. Use thermal mass construction materials in Dotito to regulate indoor temperatures. 6. Install 20,000L water storage tanks at the district hospital 7. Install solar-powered systems for water supply and cooling at all three sites. 8. Establish a demonstration site at Dotito to showcase heat-resilient infrastructure, mitigation and adaptation strategies. 9. Deploy solar-powered energy-efficient lighting, refrigeration systems, and low-global-warming refrigerants and inhalers 10. SOP guidance on heat-resilient building standards ventilation protocols and energy-efficient procurement through SOPs and collaboration with ministries	1. Lack of air conditioning in ANC facilities 2. Lack of shaded areas for HCWs to take breaks during extreme heat 3. High indoor temperatures 4. High indoor temperatures due to non-reflection of heat in uncoated roofs 5. High indoor temperatures 6. Dehydration; Lack consistent water supply 7. Lack of heat management; Dehydration ; Lack consistent water supply 8. Lack of heat management ; Lack of built environment that ensures proper heat management and indoor temperatures 9. High carbon emissions due to refrigerants ; High usage/dependence on energy inefficient lighting ; High carbon emissions due to refrigerants and high energy consuming equipment 10. High indoor temperatures
Nature based	1. Plant indigenous, fast-growing, drought-resistant trees in high-traffic areas near wards to provide shade 2. Develop facility guidance to support tree planting	1. Lack of trees and vegetation increase heat islands in facilities 2. Lack of tree shading for HCWs

6 How will the intervention be implemented

The intervention will be implemented in the same facilities as for the baseline study. These facilities include: Mamelodi Regional Hospital and Stanza Bopape Community Health Centre in South Africa, Mt Darwin District Hospital, Chitse Rural Healthcare Clinic and Dotito Rural Healthcare Clinic in Zimbabwe. An additional facility will be added in South Africa because it participates in the mitigation activities: Laudium CHC. The HIGH Horizons team directly implements some components, and others leverage collaboration with other organisations active in the environmental or occupational health fields, such as the City of Tshwane initiative for planting trees in South Africa or the Pfura Rural District Council in Zimbabwe. The beneficiaries for the intervention are also the same as for the baseline study, namely healthcare workers at all levels in the maternal and newborn health related departments in the facilities where HIGH Horizons works. The main target group for the intervention are HCWs and other staff active in maternity units and neonatal wards in the selected facilities, as for the baseline study. In South Africa, HIGH Horizons health facility adaptation interventions will be implemented in different geographical areas compared to the HAPI adaptation interventions which focus on pregnant and postpartum women's individual-, household-, community- and facility-level adaptation interventions in the west of Tshwane. In Zimbabwe, the HIGH Horizons and HAPI interventions will be implemented in the same geographical areas but are methodologically designed to minimise contamination despite their overlapping operational regions. HAPI focuses on pregnant and postpartum women and HIGH Horizons on HCWs —and implements interventions across individual, household, community, while HIGH Horizons operates at facility. Spatial segregation of activities, tailored communication strategies, and population-specific resource allocation further minimise the risk of contamination between HAPI and HIGH Horizons.

Table 6. Differences between HAPI and HIGH Horizons interventions in Zimbabwe

Intervention characteristics	HAPI: Heat Adaptation for Pregnant and Postpartum Individuals	HIGH Horizons: Health Facility Adaptation for Heat
Target Population	Pregnant and postpartum women (enrolled at 26–30 weeks' gestation)	HCWs in health facilities
Geographic Scope	Catchment areas of facilities in the district	Health facilities, specifically Mt Darwin district hospital, Chitse, Dotito
Intervention Levels	Individual, household, community, community health systems, and policy levels	Individual (HCWs), facility, and policy-level health facility guidance
Thermal Monitoring	Community dwellings in the catchment areas	Focuses on thermal conditions in the 3 health facilities
Heat Early Warning Systems	Targeted at individuals (pregnant and postpartum women)	Targeted at individual HCWs
Personal Cooling Kits	Provided to pregnant/postpartum women, CHWs	Provided to HCWs
Tree Planting Activities	Conducted at the community level	Conducted at health facilities
Awareness Activities	Radio-based shared heat messages for the community and EWS for individual women	Focuses on facility-based awareness and individual HCW capacity building
Policy Guidance	Focuses on integrating heat adaptation into maternal and child health policies	Focuses on health facility guidance and SOPs to improve heat resilience
Intervention Locations	Community-level interventions	Facility-level interventions
Cooling Infrastructure	Household-based adaptations	Facility-based interventions
Behavioural Interventions	Pregnant/postpartum women	HCW

Contamination will be minimised through the following:

- *Role segregation*: HAPI's engagement with CHWs and EHTs focuses on outreach, community mobilisation, and home-based support, while HIGH Horizons works with HCWs in their professional capacities within clinical and facility settings.
- *Separate intervention goals*: HAPI's focus is on improving maternal and neonatal health outcomes through individual-, community- and household-level adaptation interventions, while HIGH Horizons aims to enhance facility-based heat resilience for HCWs and facility operations, in order to improve ultimately

maternal and newborn health. While both studies might involve heat adaptation policy and political engagement, HAPI focuses on integrating maternal and child health heat adaptation policies, whereas HIGH Horizons addresses facility-specific SOPs to address occupational heat stress for HCWs.

- *Distinct operational domains*: The community-level heat adaptation messaging in HAPI (e.g., radio campaigns and public outreach) is designed to target general awareness, whereas HIGH Horizons focuses on facility-specific protocols and adaptations targeting HCWs and management teams.
- *Differentiated intervention levels:* HAPI implements interventions at individual (pregnant/postpartum women), household, community, and policy levels. HIGH Horizons, operates at the level of individual HCWs, health facilities, and local policy domains specifically related to the district health executive.

7 Objective of the evaluation of the climate adaptation intervention

Our objective is to evaluate the effectiveness and cost-consequences of co-designed facility-based climate adaptation-mitigation interventions in alleviating the heat impacts on health workers' wellbeing, their performance, and the quality of care they provide in South Africa and Zimbabwe.

8 Evaluation design

The evaluation will use a quasi-experimental mixed-methods approach with an interrupted time-series to establish the plausibility of effectiveness (plausibility effectiveness evaluation). Plausibility effectiveness evaluation is a terminology used by Habicht et al. (1999) to refer to evaluations which answer to the following question: "*does the intervention appear to have an effect above and beyond external*

influences". This type of evaluation aims to establish impact by using control groups, but it does not involve randomisation. Of crucial importance is to control for confounding factors and document thoroughly the implementation of the interventions as well as alternative explanations to build up a case on the effectiveness of the evaluation. Documenting the implementation and alternative explanations will be achieved through a process evaluation.

Our quasi-experimental design uses historical comparison. It includes a before-after component, with baseline data collected between August 2023 and September 2024 and the endline taking place from January-February 2026 following an implementation period of 9-13 months to allow for the different components to be fully implemented and to add to each other. Almost identical survey tools and time-motion methods done prior to the intervention will be re-administered at the same time of year as the baseline (in the hottest season). This aims to measure the plausibility of the effect of the interventions on frequent self-reported and observed outcomes related to staff wellbeing, health, productivity, and the provision of quality of care. In addition, an interrupted time series analysis of routine facility and human resources statistics in Zimbabwe and possibly South Africa, controlling for time varying confounders, will strengthen our findings, by enabling us to obtain more precise quantitative estimates of the likely effect of the intervention. Our evaluation will also include a monitoring and a qualitative component to document the implementation of the intervention using basic input indicators and observations of processes, and the additional perceived benefits (or disadvantages) reported by health workers of the intervention. Outcome data collected for the baseline and endline health worker study to establish plausibility of effectiveness will contribute to the economic evaluation. We will monitor indoor temperatures before, during and after intervention and use thermal and emissions modelling to assess changes achieved by the interventions. In all components of the evaluation, we will assess any negative outcomes of our

interventions or maladaptation. As we are also conducting a mitigation intervention in the same facilities in South Africa and Zimbabwe we anticipate several health co-benefits of the mitigation activities, and will take this into account in our analyses and interpretation.

8.1 Before/after study design

As outlined above, our intention in the before-and-after component is to use nearly identical tools, particularly for the quantitative aspects, to minimize the risk of methodological discrepancies and inconsistencies in findings.

Staff who participated in the baseline study and remain employed at the same facilities to work within maternal and newborn health services will be invited to participate in the endline research. Additionally, any new staff members in these maternal and newborn health services will also be invited to join the endline survey, provide a urine sample, participate in the time-motion studies and qualitative interviews. Details of the recruitment criteria for these new staff can be found below:

- All HCWs and other staff working in the antenatal clinics, obstetric units, including labour wards, or neonatal wards or clinics.
- Nurse of any grade, including professional nurse, auxiliary nurse and neonatal nurses; Obstetrician /Gynaecologist; Paediatrician/Neonatologist; Other types of doctors (e.g., intern, medical officer); Clinical associate; Allied health workers such as physiotherapists; Other staff working in the ward, including community healthcare worker, phlebotomist, cleaning staff and porters.
- New and past participants will be enrolled in the endline assessment only if they meet ALL the following criteria: (1) be aged ≥ 18 years; (2) be working in antenatal, labour ward or neonatal ward during the period of the study; and (3) willing to provide consent to participate in the study.

8.2 Measurement of exposure to heat adaptation intervention components

Quantitative measurement of the intervention's exposure may utilise several approaches to account for differences in intervention components. Staff working in health facilities may experience varying levels of exposure to the intervention, depending on their working patterns and locations. We must consider these factors as much as possible during the analysis (see Section 8.7) and plan accordingly during data collection.

For interventions aimed at improving the hydration of providers, exposure can be assessed through self-reported data during the questionnaire survey (see Appendix 3) or measured independently using the monitoring system that tracks interactions between providers and heat champions/environmental health technicians or track water consumption by measuring the amount of water dispensed from coolers or bottles. For interventions focused on improving the thermal environment, exposure will be evaluated through either monitoring project data or the actual measurement of changes in thermal conditions to assess improvements in the built environment (see Section 8.3).

8.3 Measurement of changes in thermal conditions before and after intervention

Indoor measurement of air and globe temperatures and air humidity along with outdoor air temperature, air humidity, wind speed, and solar irradiation was done during hot and cold seasons before any heat adaptation intervention. These measurements will continue during and after implementation of the intervention to evaluate if and how much indoor thermal exposures were moderated by the intervention. The evaluation will be based on indoor-outdoor temperature differences, maximum indoor temperatures, number of hours with temperatures above a designated threshold.

8.4 Measurement of outcomes with survey with providers

The healthcare workers survey uses an electronic survey questionnaire to record a range of indicators that can be evaluated by comparing the data collected in the baseline study against the same indicators and responses in the post-intervention period. Table 7 outlines the indicators. The endline questionnaire can be found in appendix 3. The urine specific gravity test will be done at the end of the shift as the endline questionnaire is being completed. The healthcare workers will self-collect their urine sample, which will be then assessed using a refractometer. We will use the same unique identification numbers for healthcare workers who participated in the baseline to enable data linkage. New staff members will be given their own unique identification numbers.

Table 7. List of outcomes measured with survey tool

Category	Indicators
Mental Wellbeing indicators	WHO-5 wellbeing index, Perceived stress scale-4, Generalised anxiety disorder-7
Dehydration	Urine specific gravity
Productivity indicators	Average/median number of breaks, Days of absence, Daily/hourly output (binary)
Perceived environmental conditions	Air movement, Shade, Ventilation, Humidity
Perceived thermal comfort	Sweating, Thirst, Comfort, Symptoms of heat stress
Perceived work intensity	Intensity of physical activity, Self-reported time spent in a hot environment, perceived level of fatigue
Perceived impact on Performance/Quality of care	Productivity, Medical errors, Poorer communication with patients, Poorer communication with healthcare workers (HCWs), Drugs becoming less effective, Equipment failures, Increased electricity cuts, Increased workload, Adherence to guidelines, Supply chain interruptions, Heightened irritability

8.5 Measurement of outcomes with time-motion data

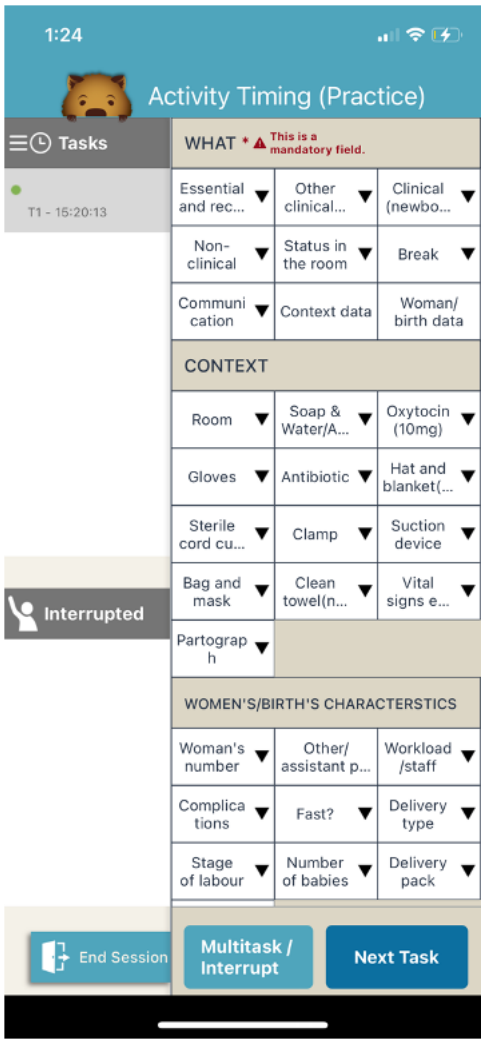
The data collection will be conducted in maternity wards within high-volume hospitals (one in Zimbabwe, one in SA) for which we already have baseline

information using the same measurement tool. A list of all eligible nurses (i.e. who are involved in the labour and delivery process) will be accessed from department heads, and recruitment activities will be conducted in labour wards. All eligible nurses who consent will be observed based on the observation schedule, their presence of the ward and their involvement in assisting labour and delivery.

All the study procedures will follow Suggested Time-motion Procedures (STAMP) (Zheng et al., 2011). We will use the Work Observation Method by Activity Timing tool (WOMBAT 3.0; Prgomet et al., 2020), a computerised tool to collect multiple dimensions of clinical work patterns (i.e. actions pursued by the health worker assisting the labour/delivery) as well as comprehensive contextual information (availability of equipment/instruments and delivery characteristics) (see Figure 1 below). When a birth attendant performs an action, the observer will record the action using a tablet. We will use continuous observation because the timing of procedures, particularly delivery itself, is typically unpredictable. We will aim to perform observations over a period of 5-10 weeks during the hotter season and at different times of the day to be able to capture patterns across different shifts and temperatures. The number of hours of observations are based on the volume of deliveries during the observation period. These observation-based methods are flexible, which is needed in the context of childbirth, for example, where there is a large amount of uncertainty (volume of clients, transition from one to two patients – mother and baby – and unpredictable complications) (Gon et al., 2020). This process will be implemented on days with forecasted high temperatures. The data will be captured by an observer using a tablet.

Preferably a clinically trained person, or junior social science researcher, will conduct the time-motion methods with supervision from an experienced social scientist.

Figure 1: Time-motion tablet interface



We provide below a few examples of the type of indicators that we will construct based on the information collected during the time-motion observation (see Table 8). These are all different aspects of processes required and prescribed by WHO/UNFPA/UNICEF guidelines on Pregnancy, Childbirth, Postpartum and Newborn Care (2015) which were contextualised for the local policies during the baseline assessment. The choice of the indicators is based on hypothesized mechanisms linked to hot temperature affecting health workers behaviour.

Table 8. List of outcomes measured with time-motion study

Outcomes	Definition
1. Break length	Time spent on a break (e.g. sitting, drinking etc)
2. Negative reactions/violence	Occurrence of: ▪ Verbal Abuse ▪ Physical Abuse ▪ Fundal pressure
3. Respectful communication	Occurrence of: ▪ Takes permission ▪ Explains procedures ▪ Ensures privacy
4. Providing appropriate advice/Supportive care	Occurrence of: ▪ Encourages hydration ▪ Encourages movement ▪ Encourages breastfeeding ▪ Encourages having a support person ▪ Giving general advice ▪ Pain management
5. Infection Prevention	Occurrence of: ▪ Hand washing ▪ Clean surfaces
6. Woman left alone in the room	Time woman in labour was left alone (period: full dilation to 1h postpartum)
7. Essential monitoring practices during labour	Completion - Number of actions performed out of 9: ▪ Temperature, FHR, pulse, duration of concentration, VE, blood pressure. ▪ Duration frequency of contraction ▪ Partograph ▪ Conditional on occurring in the first stage of labour
8. Newborn assessment	Completion - Number of actions performed out of 7: Takes newborn's temp, Takes newborn's weight, Assesses baby's colour, Assesses baby's pulse, Assesses baby's respiration, Assesses baby's reflex, Assesses baby's tone)
9. Support breastfeeding	Occurrence: ▪ Skin to skin, Support breastfeeding
10. Unrecommended practices	Occurrence: ▪ Promotion of mixed feeding/any fluids ▪ Offering artificial pacifier ▪ Amniotomy ▪ Pubic hair removal

8.6 Measurement of outcomes using routine statistics

We will use the same approach initiated during the baseline assessment. Health services and human resources daily count data have been extracted from

registers, and in South Africa, from registers and medical records, to gather additional information on performance and productivity [i.e. service utilisation data and absences (including reasons for absence in Zimbabwe)] among MNH healthcare workers during and outside periods of extreme heat. In Zimbabwe, data extraction from registers was initially retrospective, covering several years prior to the baseline assessment and has since been ongoing. In South Africa, data was initially extracted exclusively during the baseline assessment period over a span of three months during the hot season.

The initial objective (at baseline) was to provide data on context, and the initial assessment was therefore extensive. Table 9 below outlines the subset of different indicators that will be retained to assess, not only changes in context and time-variant confounders (see section 8.7), but also the potential impact of the interventions on performance or productivity between baseline and endline.

Table 9. Main proposed indicators extracted from registers

Indicator	South Africa	Zimbabwe
Healthcare utilisation	▪ Number of vaginal deliveries reported for the day ▪ Number of caesarean sections reported for the day	▪ Number of pregnant women booked for ANC at the health facility ▪ Number of women who had vaginal deliveries at the health facility ▪ Number of newborn babies receiving postnatal care on Day 7 post delivery ▪ Number of babies receiving post- natal examination at six weeks ▪ Number of women who had caesarean section at the health facility ▪ Number children hospitalised for more than 24 hours (0-4 years
Staffing	▪ Number of clinical staff allocated a shift for the day shift ▪ Number of clinical staff on leave for the day shift	▪ Number of HCW absent ▪ Number of HCW absent due to sickness ▪ Total number of staff in maternity

Indicator	South Africa	Zimbabwe
	▪ Number of clinical staff allocated a shift for the night shift ▪ Number of many clinical staff on leave for the night shift	

8.7 Analysis of survey data

We will conduct a pre-post intervention analysis, using baseline data from the hot period as a reference point. Further, the intervention implementation will offer an opportunity to conduct a dose-response analysis to measure the effect of the intervention on our outcomes. This improves investigating the causality.

The 'dose' or intensity of the exposure variable will be defined as a combination of multiple elements of the intervention or as the duration of exposure to the intervention components. We do not anticipate that all intervention elements will be implemented simultaneously across different facilities. Additionally, some components, such as interactions with heat champions in South Africa, or existing Environmental Health Technicians who will be trained as heat champions in Zimbabwe will occur repeatedly over time, and some healthcare providers will receive a higher or lower dose of the intervention depending on their working patterns and roles. Therefore, we can sub-divide the exposure into several doses or exposure intensity levels.

We will control for threats from time-varying confounders, such as staff levels and service utilisation volume (see Table 9), using variables obtained from routine data (e.g. human resources records and medical registers).

Statistical analyses will include univariate and multivariate regression techniques. For example, a linear regression model will be used to estimate the effect size by exposure dose for dehydration. Multilevel modelling will be employed where healthcare workers working in the same facility will be clustered together. We will pair the same participants in the pre- and post-intervention periods, which will

help control for within-subject variations, thereby increasing the statistical power of the analysis.

8.8 Analysis of time-motion data

We will compare the completion of the different components of care that are observed to the recorded temperatures. Data management and analyses will be conducted in STATA or R. Descriptive analyses will be used to describe the outcomes, contextual variables, and potential confounders for the relationship of the study outcomes with temperature. Observational data collected by study staff will be linked with other information available on health workers from other data collection tools in this study (e.g. health worker demographics). We will compare each of the study measures taken during the hot period at baseline and post-intervention. Multi-level model estimating the effect of external heat on quality-of-care outcomes will be run accounting for clustering at facility and individual level, and pre-specified confounders. We will also assess whether the intervention changed the relative impact of heat on quality-of-care outcomes.

8.9 Interrupted time series analysis of routine data

We will conduct an interrupted time series to assess the impact of the intervention on healthcare workers' productivity at the aggregate level (such as daily number of providers absent/present at work, daily number of caesarean conducted, daily number of antenatal care patient seen). Outcomes will be on a continuous scale and the effect of the intervention on the outcomes will be modelled using ordinary least squares regression or generalised additive models.

The association will be modelled using the following equation:

$$Y_t = \beta_0 + \beta_1 T + \beta_2 X_t + \beta_3 T X_t$$

Where Yt represents the outcomes measured on a continuous scale, T is the time elapsed since the start of the study in with the unit representing the frequency with which the observations are taken (seasonally/monthly), X_t is the dummy variable indicating the pre-intervention or the post intervention period, β_0 represents the baseline level at T = 0, β_1 is interpreted as the change in outcome associated with a time unit increase (representing the underlying pre-intervention trend), β_2 is the level change following the intervention and β_3 indicates the slope change following the intervention (using the interaction between time and intervention: TX_t).

We will control for time-varying confounders such as attendance records and shift types by including them in the model. We will assess for seasonality and long-term trend and adjust using model stratification by the calendar months and use of splines. We will assess the data for autocorrelation (if it persists after adjusting for seasonality and long-term trends) by examining the plot of residuals, the partial autocorrelation function and conducting the Breusch-Godfrey test for normally distributed data. If there is evidence of autocorrelation after adjusting for seasonality, we will instead use the autoregressive integrated moving average (ARIMA) models as they are designed to handle such data structures.

We will look for potential outliers such as extreme or unusual performance scores using Cook's distance and scatter plots as adjuncts. We will investigate the nature of any identified outliers and consider various approaches to address them such as robust regression techniques linear regression with robust standard errors and conduct sensitivity analyses with and without outliers.

We will conduct sensitivity analyses by varying model assumptions, such as different lags, types of impact models (from the equation above), and approaches for adjusting for seasonality, utilising generalised additive models to allow for non-linear relationships.

9 Process evaluation

Following MRC evaluation guidelines (Moore et al., 2015), the process evaluation will focus on: (i) contextual factors that affect the implementation of the intervention; (ii) fidelity of the implementation of the intervention as originally planned, and any adaptation along the way; (iii) responses of participants and key actors to the intervention; (iv) unexpected outcomes and consequences. This will be done using qualitative methods as well as our own monitoring system and indicators which will track the implementation of the different components of the intervention. The process evaluation is designed to track whether activities were conducted as planned, the experience of delivering them, perceptions of participants and evaluate the mechanisms of change.

9.1 Contextual factors

In documenting the context, we will cover both external and internal factors that could impact the success or failure of the intervention, including:

- **External factors** – We will consider broader influences, such as other ongoing interventions (e.g., HAPI) in the district. We will also assess changes in staffing, policies, and other quality of care, occupational or environmental factors/interventions that could shape our key outcomes (providers' wellbeing, performance and quality of care).
- **Internal, facility-specific factors** – Each facility will have its unique contextual aspects that affect the intervention such as variations in management style, human resources, and additional, concurrent interventions. They will also not receive the same intervention components. These facility-specific factors might explain differences in intervention effectiveness across locations. We will characterise each of the facility to better understand our findings.

We will document these contexts through observation, maintaining close relationships with key stakeholders, and staying attuned to any developments on the ground.

9.2 Implementation fidelity and adaptations

Assessing fidelity—whether each intervention component was delivered as intended—will be essential for determining and interpreting our effectiveness data. Given that multi-component interventions often require context-specific adjustments, particularly across different facilities, we have incorporated two "pause" points in our timeline to allow us to reflect and discuss with the facilities any adjustments to the intervention.

We will comprehensively document the implementation process, recording how and when the intervention was delivered, adjustments made and any issues related to the resources required. Fidelity will be evaluated using multiple data sources, including documentary analysis, internal monitoring data, and qualitative observations (see below). Our log frames for both South Africa and Zimbabwe provide examples of several input and output indicators, such as the number of persons recruited and trained as heat champions; the number of heat champions actively working in each facility and their start dates; records of their interactions with staff; the number of water tanks and filter machines installed, along with installation timing and duration; and the percentage of healthcare workers reporting access to cool water. These data will also help to evaluate the intensity of intervention delivery, as previously discussed (see relevant section above).

Observations in health facilities will play an important role in understanding the implementation process and fidelity. These observations will capture the implementation process, including meetings, discussions, and logistics for launching and maintaining the intervention; the frequency, consistency, and

timing of intervention adherence; the key actors involved; and any adjustments made (including timing and decision-makers).

9.3 Participants' responses to the intervention

We will investigate the *reach* of the interventions—that is, whether healthcare workers were exposed to the various intervention components, and most importantly, how this exposure occurred. We will seek to understand their perceptions of the intervention's acceptability and appropriateness, assess their views on its sustainability, and gauge whether they perceive that it is making a difference to their/their colleagues' performance and wellbeing. Additionally, we will explore the relative importance of the intervention from their perspective, considering other concurrent issues within the facility, whether related to patient care or to human resources. To gather this information, we will use self-reported data from questionnaires as well as qualitative approaches, including observations, in-depth interviews and focus group discussions.

9.4 Unintended consequences and co-benefits

In our efforts to reduce the effects of heat on healthcare providers, particularly through the installation of air conditioning, we must consider the unintended adverse consequences of our heat adaptation interventions on climate mitigation. Conversely, some of our heat adaptation strategies, such as raising awareness about the impact of heat, may encourage healthcare workers to be more sympathetic to the need for reducing carbon emissions. In addition, some of our mitigation interventions, such as planting trees for carbon offset, have a significant impact on enhancing shade and cooling the thermal environment, and are also considered in our theory of change as climate adaptation intervention.

Therefore, we will examine the trade-offs and synergistic effects—or co-benefits—of climate adaptation and mitigation interventions within health facilities. As part of our qualitative approach, we will interview authorities and healthcare providers

to understand these trade-offs and the synergistic aspects involved, with a specific focus on their impact on the work environment and quality of care.

9.5 Additional information on qualitative methods for process evaluation

Observations, interviews and focus group discussions in each country will be carried out by a social scientist and cover the four main areas of the process evaluation. Key topics will include the perceived need for an intervention to improve conditions during extreme heat events; involvement in the intervention's design and consultation process; assessment of factors that contribute to the intervention's success or challenges.

Specific attention will be paid to the participants' perception of the intervention and how it influenced their experience of heat during their workday (and possibly extending beyond the hospital to commutes; home life; etc.) regarding their performance, their wellbeing and the quality of care they provided to mothers and newborns.

The qualitative methodology is designed to be responsive to the intervention and how it is implemented. As such, further points of connection may be introduced during the 12-month fieldwork period in order to understand the process, for example additional observation periods may be introduced or key informants interviewed.

❖ Observations

To document contextual factors, fidelity, and adaptation, an observation schedule will be arranged. Over the twelve-month period of the intervention, it is anticipated that an observer visits the facility between 4 and 8 times to carry out a set of observations lasting between 1 and 4 days. These visits will be scheduled to occur during both key intervention moments (for example, internal meetings to discuss modifications to the intervention or events held related to the intervention) events

held related to the intervention) and routine practices, to document how staff work alongside the intervention and their responses to it. Observations will include informal conversations with staff and follow an observation guide. Detailed notes in the form of a daily or activity journal will be kept and transcribed for analysis.

- **In-depth interviews with healthcare providers, hospital administrators and other informants**

This same social scientist will conduct interviews with 10-20 health workers, affiliated health staff and hospital administrators in each maternity ward, preferably primarily with those who have been working in the ward during the baseline study. The exact profile and number of interviewees will be identified during data collection and will depend on whose perspectives are identified as significant to the intervention, and the principle of saturation will guide sampling. The researcher carrying out observations will also conduct the interviews, so that insights from observation can inform the line of questioning at each facility.

These interviews will take place at various points across the implementation period. A likely schedule will be that the researcher conducts 2-3 interviews during each observation visit, and that after the focus group discussion towards the end of the project 5-10 in-depth interviews (IDIs) are carried out to follow up on themes that were raised in discussions and noted during observation periods.

Interviews are expected to last between 30-60 minutes, will be audio-recorded, follow an interview guide and take place in a private setting that suits the participant. Written informed consent will be attained before the interview.

- **Focus group discussions**

A focus group discussion (FGD) will be convened near the end of the 12-month intervention period, to hear from a selection of staff at the maternity ward about their experience of the intervention. The FGD is designed to foster dialogue amongst the participants so that they can share, listen to and react and engage

with each other's perspectives. Between 8-12 participants will participate in this FGD. If the researcher decides that it is better to create two groups depending on staff profile to foster openness and reflection, there may be two FGDs. FGD(s) will be led by a facilitator, accompanied by a second facilitator and notetaker, consented to by all participants in writing, audio recorded and transcribed for analysis.

The new or revised qualitative tools are included in Appendix 4.

9.6 Analysis

Comparative analysis will be done across time (baseline and post-intervention) and across health facilities, with the nature and degree of implementation of the intervention in each setting providing a lens through which to analyse its impact and influence. Qualitative software such as MAXQDA or NVivo will be used to manage and code the data. Data will be coded thematically according to a priori themes and those that emerge during fieldwork. Analysis reports per country will be produced with an eye to informing the implementors and to serve as the basis for future scientific publications.

10 Cost component/economic evaluation

We will undertake a cost consequence analysis (CCA) of the different climate mitigation and adaptation interventions in both Zimbabwe and South Africa.

10.1 Study design for cost component

The cost analysis follows the Global Health Costing Consortium (GHCC) reference case which provides a standardised framework for global health costing studies (Vassall et al, 2018). The purpose of this costing is to estimate the total and incremental economic costs of implementing new climate adaptation and mitigation interventions. We will estimate financial and economic costs for all the

intervention components in South Africa and Zimbabwe. Costs will be estimated from the provider's perspective.

Costs from the providers' perspective will be identified and estimated using a combination approach, using both top-down and bottom-up costing methods. Data will be collected using a costing tool presented in the Appendix 5. For each intervention, specific activities and resources (ingredients) will be identified, measured and valued. These will include capital costs, such as building and equipment with a lifespan exciding one year, as well as personnel costs, and recurrent costs including supplies and consumables with a lifespan of less than one year.

10.2 Cost data analysis

We will determine the total cost of developing and implementing the different mitigation and adaptation interventions, as well as the unit cost for healthcare worker using the intervention. We will report and describe the cost breakdown and distribution of costs across intervention components.

We will consider the budget impact of scaling up the different interventions across different settings and scenario. Cost modelling work will inform on the impact of scaling up the intervention proposed. Once full costs are obtained to estimate value of all resource used for each intervention, we will investigate the budget impact of distribution across country.

10.3 Valuation of resources used

Costs will be reported in USD2024 using a 3% discount rate where costs need to be adjusted over time. For capital goods we will consider depreciation of capital costs using a discount rate of 3%, while using country specific discount rates in the sensitivity analysis.

10.4 Cost-consequence analysis

We will estimate the costs for consequences of the different adaptation and mitigation interventions. Consequences for the mitigation interventions will include changes in CO_2 emissions. These will be measured using the CARBOMICA tool. For the adaptation interventions, different consequences will be considered including uptake and usage of interventions, changes health-workers wellbeing and productivity and changes in heat strain score index. These changes will be measured from the semi-structured questionnaires.

11 Ethical considerations

Healthcare workers are familiar with the project, having participated in baseline assessments, and some were involved in the co-design of interventions.

11.1 Informed consent

We will follow the same informed consent procedures as for the baseline which are:

Before each participant is invited to continue to participate or enrolled into the study, an informed consent will be obtained in accordance with local and international ethical guidelines. The participant will be given a copy of the informed consent, and the research team will retain a copy of the informed consent. The informed consent will be attached to a participant information sheet that will provide further information about the study, with an emphasis about voluntary participation. Health workers will be asked to provide separate written informed consent for each of the study components. Participants can decline participation in all or some components of the study without disadvantages or repercussions. They can withdraw their consent for study participation at any time.

There are several important ethical considerations regarding the participant observation activity. Principally, it is important that people know that they are being observed, what the observer is doing in the area, what the purpose of their activity is, and how the observation data will be used. We will do this in several ways, tailored to each group being observed.

Firstly, for healthcare workers stationed in the wards where the observation is taking place, we will inform them collectively (in a meeting) of the purpose of this activity and what it will entail. We will explain that no personal identifying information will be collected on the individuals who are observed. Healthcare workers will be given a chance to ask questions and express any concerns about the activity, and these will be addressed by the study team in the same meeting. It will be explained that participating in the observation activity is voluntary, and refusal to participate will have no repercussions for the individuals concerned. Written informed consent will then be obtained individually from all healthcare workers who agree to be observed. This consent will apply to the entire period of observation in each facility.

Secondly, for the wider pool of people who may enter the field of observation, including pregnant intrapartum or postpartum women, visitors, and other facility staff, the research team will not seek written informed consent. This is not logistically feasible as it would necessitate a very time-consuming practice of approaching every person to obtain informed consent, even if they are merely passing through the area, and in many cases these wards and waiting rooms are very busy. Nonetheless, it is still important that people know they may be observed in these spaces. We will develop a simple, easy-to-read poster that explains the process of observation, what kind of data are being collected through observation in this study, and how the privacy and confidentiality of those being observed will be maintained. The poster will also make it clear that individuals may opt out of being observed, if they so wish, and they may do so by bringing it to the attention

of the observer, who will be wearing an identifying badge. No information about the actions or words of individuals who have opted out of the activity will then be recorded by the observer.

Thirdly for the time-motion study, while the person who is observed is the healthcare worker, documenting the timing of their activities, as they go about providing care during labour, we will only carry out this fieldwork with the explicit permission of the women themselves who may object to have a third party in the delivery room.

Permission from management has already been obtained to install measurement devices in selected rooms in the facilities.

11.2 Risks

There are minimal risks associated with participation in this evaluation. The are risks around data protection which is mitigated through data security measures to ensure participant confidentiality detailed further in the data management plan.

Participating may be psychologically uncomfortable and stressful for some participants as some questions on their wellbeing and productivity may be considered personal. Participants may also be uncomfortable with the experience of being observed. These risks will be explained to the participants.

The collection of biomarkers (urine) and data on stress and anxiety may reveal staff who might benefit from clinical attention. Staff will have access to their results if they wish to have these. Staff who have gone above certain thresholds suggesting that they might benefit from clinical or mental health services will be informed and offered referral to occupational health in their facility/district, after discussion with the lead clinical researcher in each setting. The urine testing is unlikely to lead to physical discomfort.

Fieldwork will take minimal time to complete to avoid any interruptions to patient care in the healthcare facility.

11.3 Anonymity, confidentiality and safety

All healthcare worker participants have already received or will receive if they are new staff a unique identification number.

All data will be anonymised and pseudonymised.

The participants' confidentiality will be protected in accordance with the General Data Protection Regulation.

In each country, a file linking names and unique identification number will be kept electronically using encryption and password protection and be accessible only to the selected project staff with different levels or access rights. If participants have been involved in different components of the study, their study identification will remain the same to allow researchers to link data across components if required. No information that has personal identifiers will be shared with the participant's employer or any external partners.

The evaluation results will be published for scientific purposes, but anonymity will be protected at all times, and we will ensure that as far as possible, no results can be linked to specific individuals.

12 Data management

Data management will be carried out in accordance with the HIGH Horizons Data Management Plan (Luchters and Kerstens, 2024, Zenodo) and related updates. In South Africa, processes will adhere to the regulatory frameworks of the South African Protection of Personal Information Act (POPIA) and in Zimbabwe, process will adhere to the Zimbabwe's Data Protection Act [Chapter 11:22] (2021), ensuring compliance with jurisdictional data sovereignty requirements and cross-border data transfer restrictions. In Zimbabwe, explicit written consent is mandatory for processing sensitive data, including health, and bio information (Sections 11(1)

and 12(1)), with stringent measures enforced to safeguard data integrity, confidentiality, and protection against unauthorised access (Section 18(1)). Furthermore, research involving human participants will be subject to the oversight of the Medical Research Council of Zimbabwe (MRCZ), to ensure alignment with both legal mandates and ethical principles governing data collection, storage, and usage.

In accordance with the HIGH Horizons Data Management Plan, our data management has the following characteristics, which is common to all components, unless stated otherwise:

- As for the baseline, each new participant will have a unique identification number, which will be the same in each quantitative and qualitative sub-study (possible example: country code, followed by facility code, followed individual number ranging from 1-200). This allows us to link data of individuals who participate in multiple components. All hard copies of patient identifying data will be stored in a locked cabinet. Baseline and endline data will also be linked using the unique identification number.
- All of the quantitative data will be collected electronically on a tablet or smartphone, and backed up as per each partner's Standard Operating Procedures.
- All of the data will be analysed electronically.
- Quantitative data will be fully anonymised. Pseudonyms will be used in reporting of the qualitative research, for example, when participant quotes are used.
- Each partner will store the data collected on its institutional network, on a secure server protected by two-level authentication.
- Data storage on local devices will be avoided at all times. If storing a copy of the data on local machines is necessary (e.g., the study statistician), then the data will be password protected and only include deidentified data.

- Access to data will be restricted to a limited number of study staff that have appropriate training to manage data.
- Data will not be sent by email but transferred safely using encryption or secure tools like Filesender.
- As most data are health-related, personal and sensitive, we cannot provide full open access to our datasets. As such, the datasets will not be entirely openly available, and conditions will be attached to access the data (i.e., restricted access). We will take actions to limit the restrictions, i.e., to have our datasets "as open as possible, as closed as needed", by pseudonymisation/anonymisation of personal data where possible and by agreeing on a limited embargo.
- Non-personal data (e.g., environmental or ambient temperature data) will be made openly available unless sourced from a commercial source.
- Data ownership will be with the country teams in South Africa and Zimbabwe. (and Sweden for the baseline data).
- Fully anonymized data will be shared with LSHTM for cross-country analysis purposes.
- All data will be stored in accordance with country legislation and recognised data security standards. Specifically, data will be archived once study results are analysed for a minimum of six years in South Africa and 10 years in Zimbabwe.

Further details of data management for some specific study components are provided below.

12.1 Qualitative enquiry datasets

Qualitative data, such as from the process evaluation component, will be managed in the first instance by each country partner in line with national data protection laws and research ethics guidelines. Interview recordings will be transcribed and quality checked and securely stored.

In South Africa, recordings will be securely stored for two years after research findings are published, or for minimally six years if the findings are not published, in accordance with research ethics protocols, and then destroyed. Researchers will type up fieldnotes, producing a pseudonymised version that can be shared only within the research team.

In Zimbabwe, qualitative data management will align with international best practices and comply with the Zimbabwe Data Protection Act [Chapter 11:22] (2021) and Medical Research Council of Zimbabwe (MRCZ) guidelines. Audio recordings of interviews and focus groups will be captured using secure devices with participants' written consent, as required by Sections 11(1) and 12(1) of the Act. Recordings will be transferred to encrypted servers within 72 hours, with temporary local storage deleted immediately. Fieldnotes will be transcribed within 120 hours, pseudonymised, and verified for accuracy. Audio recordings will be retained for two years post-publication or six years if unpublished. Transcripts and fieldnotes will be kept for up to five years post-publication. After the retention period, audio files will be securely deleted, and printed materials shredded. Data sharing with external partners will require additional consent and comply with Section 28(1) of the Act, ensuring equivalent data protection standards in recipient countries. Audio files, transcripts, and fieldnotes will be securely stored on institutional servers with two-factor authentication.

Files will be analysed by each of the country teams of researchers, using qualitative software such as MAXQDA or NVivo, and pseudonymised data will be securely transferred to the LSHTM server using encrypted tools like FileSender for cross-country analysis with access restricted to trained researchers. Participant consent for sharing of pseudonymised outputs with project partners will be sought during data collection. Participants will be asked to consent for further follow-up and for keeping their contact details securely on countries' respective servers for the duration of the project.

12.2 Time-motion assessment

Data on health workers' behaviour captured using computerised direct observation methods consist of electronic data linked to individual providers with unique identification numbers. Data will be rendered fully anonymous during analyses. Data generated in the course of research will be kept securely in electronic format as appropriate, in accordance with good practice in the storage of primary data, record keeping, ethical issues, GDPR legislation as well as POPIA in South Africa and Cyber and Data Protection Act in Zimbabwe. Identifiable material will be stored separately from any observation notes and kept locked in drawers or protected computers, for example, where there are hard copy informed consent forms.

12.3 Survey of healthcare workers

Quantitative data from the structured interview survey will be collected electronically using an electronic platform such as REDCap or ODK. Once collected, electronic data will be kept in a secure password protected server at each site. The survey is self-administered; hence participants will enter all data directly onto a secure web data entry form. Data logic and validation checks will be configured into the survey, where possible, so that participants will receive warning messages at time of entry of inconsistent, incomplete or invalid data, to ensure quality and completeness of information. The self-reported data will not be queried for clarification or correction by study staff after electronic submission of the survey.

Each site will anonymise and clean data to ensure the confidentiality of the participants and data quality. Each site will use the data for their in-country analysis. Files will subsequently be securely transferred to the LSHTM server for cross-country analysis. Participant consent for sharing of anonymised outputs will be sought during data collection.

12.4 Extraction of routine data

These are daily/weekly/monthly counts of aggregated data on number of events (i.e., caesarean section). Data will be extracted from registers and similar facility records by study personnel, using a computer or tablet, and entered into the same tools as described in section 12.3. Security considerations, and logic and validation checks, will be similarly implemented. The list of variables is defined in Section 8.6.

12.5 Assessment of thermal exposure

Data on thermal exposure consist of time-series of temperatures and air humidities representing the entire measurement period. Data will be retrieved from the instrumentation in CSV format for further processing and analysis in statistical software such as Stata or R. Data is stored on the logging devices and will be read out regularly by researchers visiting the facilities. Combined with data on clothing insulation and activity, indices for the assessment of the thermal exposure will be calculated and interpreted based on guideline values provided in the relevant standards.

12.6 Cost component/economic evaluation

Data collected for the cost component will be kept strictly confidential, using the same security measures described earlier in this section. Any documentation received from stakeholders for capturing will not be kept any longer than is necessary for analysis, and will be returned or destroyed as applicable.

13 Timeline

The identification of ideal areas for planting trees was one of the first activities in South Africa, because 100 indigenous trees were donated and planted in September 2024. Further site preparations and procurement for the

implementation of the selected adaptation interventions in South Africa started since December 2024.

In Zimbabwe the preparatory work starts in March 2025, with implementation protocol training and engagement with chiefs, local governance and public work authorities to get permissions to do retrofitting of the built environment and to inform them about the further implementation steps.

Implementation of the adaptation interventions will run through the cold and hot seasons in South Africa and Zimbabwe till January-February 2026 when the endline assessment of the health worker study and the evaluation of the cost-effectiveness of the adaptation interventions are planned.

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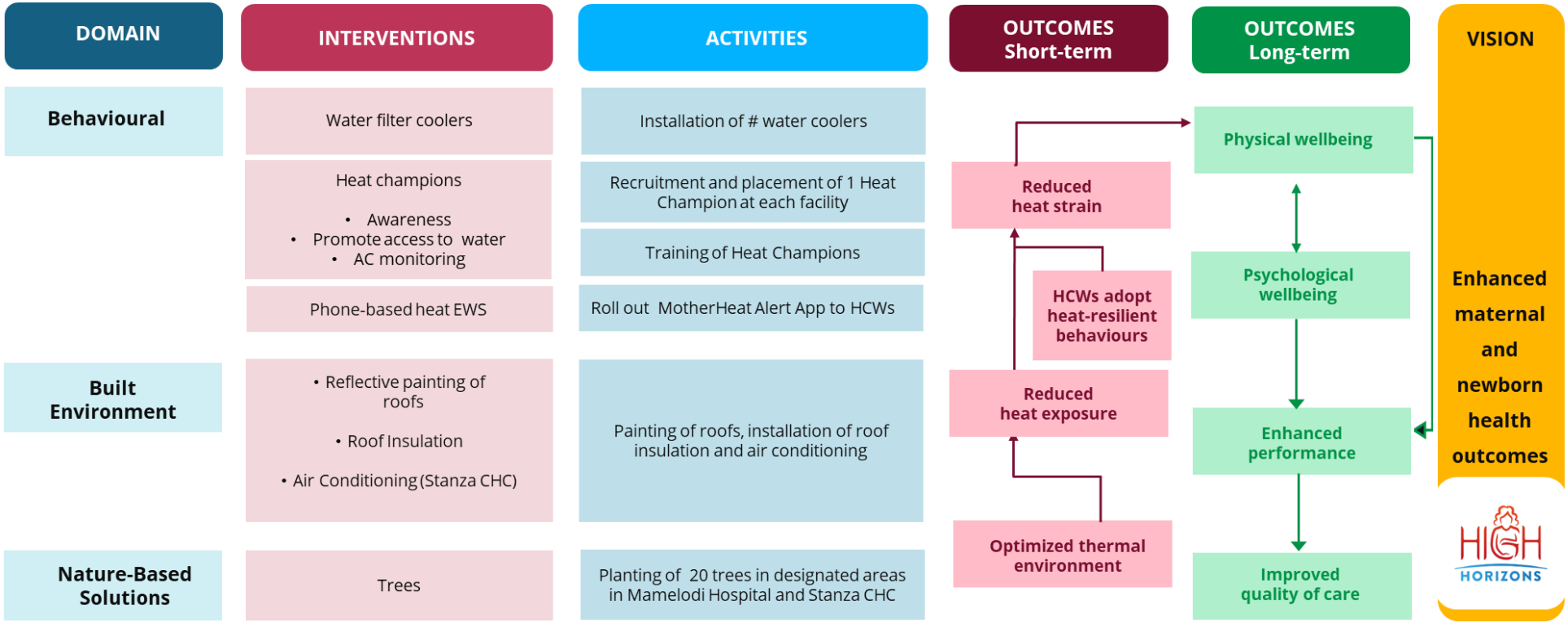
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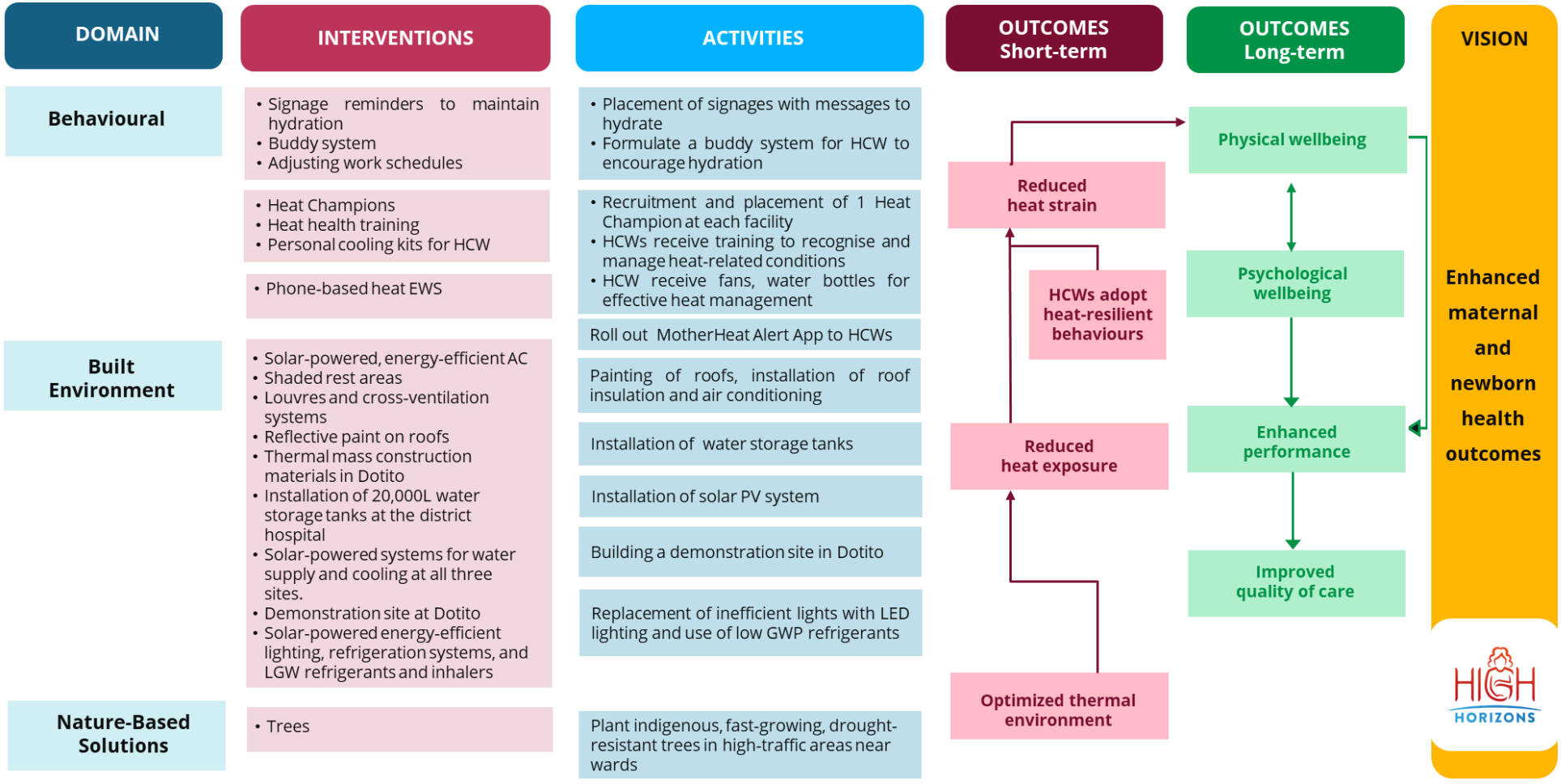
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15 Appendices

15.1 Appendix 1: Theory of change for adaptation interventions in South Africa



15.2 Appendix 2: Theory of change for adaptation interventions in Zimbabwe



15.3 Appendix 3: Endline questionnaire for health workers

Endline_HIGH HORIZONS Health Worker Survey Questionnaire

Name of country

- ◯ Zimbabwe
- ◯ South Africa
- ◯ Sweden

Name of facility

Participant ID Label

Our records show that you have participated in the baseline. Can you confirm that?

Thank you for accepting to take part in the HIGH Horizon Healthcare Worker Study. The questionnaire will take approximately xx minutes to complete and can be done in more than one sitting preferably on the same day, towards the end of your shift. Your responses are important.

Demographic and work variables

Q2_1. How old are you?

_____

Q2_2. What is your sex?

- ◯ Female
- ◯ Male
- ◯ Prefer to self-describe
- ◯ Prefer not to say

Q2_3. What is your highest level of education?

- ◯ No schooling
- ◯ Primary school
- ◯ Some secondary
- ◯ Secondary school (can include O levels, A levels and matric)
- ◯ National Diploma
- ◯ Advanced Diploma,
- ◯ Undergraduate degree
- ◯ Post-graduate degree
- ◯ Prefer not to say

Q2_4. What is your staff position in this hospital?

- ◯ Nurse of any grade
- ◯ Midwife
- ◯ Obstetrician/Gynaecologist
- ◯ Paediatrician or Neonatologist
- ◯ Anaesthesiologist
- ◯ Other types of doctor (e.g., intern, medical officer)
- ◯ Clinical associate
- ◯ Allied health workers such as physiotherapists
- ◯ Community health worker
- ◯ Cleaner
- ◯ Porter
- ◯ Other, please specify

Q2_4. Specify position

_____

» Q2_5. How long have you worked as a healthcare worker?

Years

_____

Months

_____

» Q2_6. How long have you worked in this facility?

Years

_____

Months

_____

» Q2_7. How long have you worked in your most recent role (e.g. a health facility manager; senior midwife etc)?

Years

_____

Months

_____

Q2_8. Do you work part-time or full-time?

- ◯ Part-time
- ◯ Full-time
- ◯ Locum/Self-employed/per shift

Q2_9. If you have a contract, how many hours per week are you contracted to work at this facility?

_____

Q2_10. In which service/department did you spend most of your time over the past 7 days?

Select all that apply

- [ ] Antenatal clinic
- [ ] Antenatal ward
- [ ] Labour and delivery room
- [ ] Surgical theatre
- [ ] Postnatal ward
- [ ] Neonatal ward
- [ ] Other outpatient
- [ ] Other wards
- [ ] Outreach
- [ ] Other/ not in facility

Q2_11. How long is your current/ was your last shift?

In hours

_____

Q2_12. How many breaks pauses during your shift work have you taken today, of 5 minutes or more?

Please provide the number of breaks, not the length. For example, if you have taken nearly three breaks, then enter 3

_____

Q2_13. During the last two weeks, how many of your working days were you absent (for any reason, including sick leave) from the facility?

_____

Q2_14. Overall, on a scale of 1-10 with 10 being the highest level, how motivated overall do you feel to work in this facility?

1 10

Health status in general

We would like to know about your health status.

Q3_1. Would you say that today your health in general is: Excellent, very good, good, fair or poor?

- ◯ Excellent
- ◯ Very good
- ◯ Good
- ◯ Fair
- ◯ Poor
- ◯ Don't know

Q3_2. Do you currently have any of the following illnesses (as diagnosed by a healthcare professional)?

Select all that apply

- [ ] Hypertension
- [ ] Diabetes
- [ ] Dyslipidaemia (high cholesterol or triglycerides)
- [ ] Asthma/COPD
- [ ] TB
- [ ] Autoimmune disease, e.g., Lupus, rheumatoid arthritis
- [ ] Osteoarthritis
- [ ] Heart disease
- [ ] Stroke
- [ ] Epilepsy
- [ ] Depression/anxiety
- [ ] Other mental illness
- [ ] HIV
- [ ] Other (please specify)
- [ ] None of the above

Q3_2 Specify other illness

_____

Q3_3. If you are a woman, are you pregnant or were pregnant in the past 12 months?

- ◯ Yes, I am currently pregnant
- ◯ Yes, pregnant in the past 12 months
- ◯ No
- ◯ Not applicable
- ◯ Don't know

Your wellbeing: WHO Well-being Index for Health Workers tool

For each of the five statements, please indicate whether in the last two weeks you have been feeling this way all of the time, most of the time, more than half of the time, less than half of the time, some of the time, or never.

How often	All of the time	Most of the time	More than half of the time	Less than half of the time	Some of the time	Never
Q4_1. In the past two weeks, I have felt cheerful and in good spirits ...	○	○	○	○	○	○
Q4_2. In the past two weeks, I have felt calm and relaxed	○	○	○	○	○	○
Q4_3. In the past two weeks, I have felt active and vigorous...	○	○	○	○	○	○
Q4_4. In the past two weeks, I woke up feeling fresh and rested....	○	○	○	○	○	○
Q4_5. In the past two weeks, my daily life has been filled with things that interest me...	○	○	○	○	○	○

Your stress level: PSS-4 (Perceived Stress Scale 4)

The questions in this scale ask about your feelings and thoughts during the last month. **In each case, you will be asked to indicate how often you felt or thought a certain way. Although some of the questions are similar, there are differences between them and you should treat each one as a separate question.**

	Never	Almost never	Sometimes	Fairly often	Very often
Q5_1. In the last month, how often have you felt that you were unable to control the important things in your life?	○	○	○	○	○
Q5_2. In the last month, how often have you felt confident about your ability to handle your personal problems?	○	○	○	○	○
Q5_3. In the last month, how often have you felt that things were going your way?	○	○	○	○	○

Q5_4. In the last month, how often have you felt difficulties were piling up so high that you could not overcome them?	◯	◯	◯	◯	◯

Your anxiety level: GAD-7 (Generalised Anxiety Disorder)

How often have you been bothered by the following over the past 2 weeks?

	Not at all	Several days	More than half of the days	Nearly every day
Q6_1. Feeling nervous, anxious, or on edge	◯	◯	◯	◯
Q6_2. Not being able to stop or control worrying	◯	◯	◯	◯
Q6_3. Worrying too much about different things	◯	◯	◯	◯
Q6_4. Trouble relaxing	◯	◯	◯	◯
Q6_5. Being so restless that it's hard to sit still	◯	◯	◯	◯
Q6_6. Being easily annoyed or irritable	◯	◯	◯	◯
Q6_7. Being afraid as if something awful might happen	◯	◯	◯	◯

Adapted heat strain score index

Q7_1. During your work, the level of physical activity you engage in can be described as:

- ◯ Very light similar to resting
- ◯ Light
- ◯ Moderate
- ◯ Hard
- ◯ Very hard

Q7_2. The air movement in my workplace can best be described as:

Air movement here refers to any form of air current or draughts circulating in the room. More air movement improves heat impact on individuals

- ◯ No air movement
- ◯ Light air movement
- ◯ Moderate air movement
- ◯ Strong air movement
- ◯ Very strong air movement

Q7_3. The sunlight in your workstation can be described as:

- ◯ Full shadow
- ◯ Sun and shadow or mild light
- ◯ Full sun or intense light

Q7_4. The ventilation (through fans and conditioners or natural through windows) in my workplace can best be described as:

Ventilation can be mechanical (e.g. fans) or natural (through windows and doors.

- ◯ Good and effective ventilation
- ◯ Appropriate ventilation no additional ventilation is required
- ◯ Inadequate ventilation
- ◯ no ventilation

Q7_5. How much of your time is spent in a hot environment, in a shift?

- ◯ Less than 2 hours
- ◯ 2-4 hours
- ◯ 4–6 hours
- ◯ More than 6 hours

Q7_6. Thinking about the hottest day in the past week, how did the temperature in your workplace feel?

- ◯ Cool
- ◯ Temperate
- ◯ Warm
- ◯ Hot
- ◯ Very hot

Q7_7. Thinking about the hottest day in the past week, how did the air humidity feel in your workplace?

- ◯ Dry
- ◯ Slightly humid
- ◯ Moderately humid
- ◯ Very humid
- ◯ Extremely humid

Q7_8. Thinking about the hottest day in the past week, how much sweating did you experience on the hottest day of the past week on a scale of 0-5? Zero represents no sweat, and 4 severe sweat.

0 5

Q7_9. Thinking about the hottest day in the past week, the level of fatigue that you experience at work can be described as:

- ◯ Not tired at all
- ◯ A little tired
- ◯ Tired
- ◯ Exhausted
- ◯ So exhausted I wanted to have a break

Q7_10. Thinking about the hottest day in the past week, how thirsty were you when you were at work?

- ◯ I don't get thirsty
- ◯ I get a little thirsty
- ◯ I get thirsty
- ◯ I get very thirsty
- ◯ I get so thirsty that my mouth and throat get dry and they can't be wet with saliva

Q7_11. Thinking about the hottest day in the past week, how comfortable were you at work?

- ◯ Comfortable
- ◯ Slightly uncomfortable
- ◯ Uncomfortable
- ◯ Very uncomfortable
- ◯ So uncomfortable I want to quit my job

Q7_12. Thinking about the hottest day in the past week , did you feel that you suffered from symptoms of heat stress (Thirst, sweating, thermal discomfort, headaches, feeling tired, other symptoms?)

- ◯ Yes
- ◯ No
- ◯ Unsure

Q7_13. Have you ever experienced heat stroke in your workplace?

Heat stroke is define as "a core body temperature that rises above 40 degrees C, accompanied by hot dry skin and central nervous system abnormalities, such as delirium, convulsions, or coma"

- ◯ Yes
- ◯ No
- ◯ Unsure

Perceptions of heat's influence on quality of care. (Only for hot season)

Q8_1. During the hot season, does heat exposure reduce your daily or hourly work input?

- ◯ Yes
- ◯ No
- ◯ Don't know

» **Q8_2. When you think about the last heat wave that occurred in your area, have you observed any of the following during the heatwaves?**

Frequency	Never	Rarely	Often	Every time	Unsure
Q8_2-1. Reduction of productivity per person?	○	○	○	○	○
Q8_2-2. Increase in the number of medical errors?	○	○	○	○	○
Q8_2-3. Poorer communication with patients?	○	○	○	○	○
Q8_2-4. Poorer communication between health care workers?	○	○	○	○	○
Q8_2-5. Patients in less clean environment?	○	○	○	○	○
Q8_2-6. Drugs (e.g. oxytocin) becoming less effective?	○	○	○	○	○
Q8_2-7. Equipment faultier?	○	○	○	○	○
Q8_2-8. Increased electricity cuts?	○	○	○	○	○
Q8_2-9. Increased clinical workload?	○	○	○	○	○
Q8_2-10. Providers less likely to follow guidelines?	○	○	○	○	○
Q8_2-11. Supply chain interruption	○	○	○	○	○
Q8_2-12. Heightened Irritability	○	○	○	○	○

Current heat adaptations. (Only for hot season questionnaire)

» **Q9_1. Could you please rate the availability of the following heat adaptation measures in your clinic/ward?**

	Never	Rarely	Sometimes	Often	Always
Q9_1-1. Availability of shading devices to reduce direct sunlight? *i.e. blinds, curtains, overhangs, tents, veranda etc*	○	○	○	○	○
Q9_1-2. The availability of water for staff or patients in the wards?	○	○	○	○	○
Q9_1-3. The availability of cold refrigerated water for staff or patients in the wards?	○	○	○	○	○

Q9_1-4. The availability of functioning air conditioning?	◯	◯	◯	◯	◯
Q9_1-5. The availability of fans?	◯	◯	◯	◯	◯

Q9_1-6. The availability of heat champions?

- ◯ Never
- ◯ Rarely
- ◯ Sometimes
- ◯ Often
- ◯ Always

Q9_1-7. The availability of cooling vests?

- ◯ Never
- ◯ Rarely
- ◯ Sometimes
- ◯ Often
- ◯ Always

Q9_1-8. The presence of roof insulation?

- ◯ No
- ◯ Yes

Q9_1-9. The presence of reflective roof painting?

- ◯ No
- ◯ Yes

Q9_1-10. The availability of occupational health safety compliance assessments?

- ◯ Never
- ◯ Rarely
- ◯ Sometimes
- ◯ Often
- ◯ Always

Q9_1-11. The amount of trees in the immediate proximity (<5 meters) outside the ward/clinic?

- ◯ Many trees all the way around
- ◯ Some trees in certain areas
- ◯ Few trees
- ◯ No trees

Q9_1_12. The availability of spaces to cool down for yourself or your patients when it is too hot?

- ◯ Many spaces that are always available
- ◯ Many spaces that are often available
- ◯ Few spaces that often available
- ◯ Few spaces that are rarely available
- ◯ No spaces available

15.4 Appendix 4: Endline qualitative tools

4a: Interview guide for health workers, hospital staff and key informants

Please note: this guide will be adapted according to the profile of the interview participant and the nature of the intervention co-designed with their facility. As the interviews will take place at different points in the intervention, at times, only a selection of the question topics may be asked.

Profile of participant

- Role at facility, where they are from, number of years working there

Personal heat experience (icebreaker, as necessary)

- What to you is the ideal temperature range to work and live your day-to-day life? How so? Please explain.
    - *Probe as to whether these ideal temperature ranges vary for recreational activities and for work, and if so, why?*
- Do you think extreme heat impacts your day-to-day life? If so, how?
    - At home
    - Commuting to work
    - At work

Co-design process

- Were you involved in the co-design process? In which capacity?
- Can you describe to me how the health facility heat intervention development was undertaken at this facility? (Prompts: who participated, steps involved,
- Which factors were taken into consideration during intervention development?
- Do you think the right people were involved in the development? (How, how not, who should have been there/not been there?)
- Do you think people's voices and opinions were heard and considered? Please give examples.
- How were decisions made about what to include and not include in the intervention? Please give examples
  What sources, evidence or experiences were consulted in the development of the intervention plan?
- Were you satisfied with the process? Why, why not?

Implementation

- Do you think that the interventions are being implemented in the way they were intended? In which ways?
- What determines whether the intervention is implemented fully?

Experience of intervention

- Overall, do you think that the intervention is making a difference? In which ways?

Let us discuss the different components to the intervention.

- What are your thoughts on each one, thinking about how well they are accepted by the team, how feasible they are to integrate into day-to-day practices, whether they were implemented fully and faithfully, and whether they make a difference in alleviating heat strain.
  [Interviewer lists each intervention one by one, so as to get feedback on each component.]

Impact

- Do you think that the interventions have changed your
  - Performance?
  - Well-being (physical & mental)?
  - Services provided?

- For each one, please explain, and give examples, if possible. [*Interviewer, try to disentangle if there are certain components that provide more benefits for the three than others.]*

- Do you sense that patients have been able to notice a difference since the start of the intervention? If so, how, in which ways?

- Do you think the results of the intervention are important to patients?

- What do you think should be done differently next time?

- Having gone through this process, do you have other suggestions for interventions that you think would work in this setting? Which ones?

Contextual factors

- Which contextual factors (at the facility, and beyond) do you think supported the interventions in being carried out?
- Which factors hindered the intervention's implementation?

Sustainability

- How likely do you think it is that the changes made continue to be instituted 3 months from now? 3 years from now?
- Do you think this program is sustainable? Should it be continued?

- Have you had experience from other interventions in your work as a [health worker] that you feel have had a positive impact on your work practices? Can you name the intervention/program and what aspects worked well? Why do you think those worked well?

- What do you think about the attention being given to heat stress now at your health facility? Do you think the attention is well-placed?

Trade-offs

- Are there trade-offs and synergistic effects (or co-benefits) of climate adaptation and mitigation interventions being carried out at this health facility?
- What are they, can you explain? *[Prompt on health worker well-being, performance, quality of care]*

Concluding thoughts

- What are your recommendations to improve the program your facility has undergone to improve heat adaptation?
- Do you think that even just the attention and activity around the subject is beneficial, to raise awareness?
- Is there anything else you would like to add about the interventions or your experience of heat?

4b: Observation guide at health facilities

Introductory note for the researcher

A researcher will carry out participant observation in health facilities according to a schedule designed to follow and respond to the intervention period at each focal facility. Staff members will also be informed of the researchers' presence and purpose, and implied consent is to be obtained. This will likely involve an introductory group meeting explaining the research and the nature of the observation, or a flyer explaining the research and its principles of anonymity and confidentiality.

The researcher will keep a detailed written and/or oral field journal (to be elaborated on after observation hours). This is a guide with suggestions of what to pay attention to and document along the lines of the process evaluation, with an eye to keeping the study research questions and objectives at the forefront of participant observation. Remember that this will be comparative work as the facility and practices will be observed in comparison to the baseline research in hotter and cooler seasons.

Observation guide

This guide outlines key areas for observation to capture the process of an intervention in a health facility. Observations should focus on the contextual factors, fidelity, participant and actor responses, and any unexpected outcomes of the intervention.

Contextual factors affecting implementation

- **Broader facility factors, including infrastructure and resources**
    - Note broader influences, such as other ongoing interventions (e.g., HAPI) in the district.
    - Assess changes in staffing, policies, and other quality of care, occupational of environmental factors /interventions that could shape the key outcomes (providers' wellbeing, performance and quality of care).
    - Note any competing activities or space sharing within the facility that might affect the intervention's operation.
    - Observe the availability and functionality of space, equipment, and supplies specifically used for the intervention.

- **Staffing and workload**
    - Observe staffing levels and the overall workload, noting any periods of staff shortages or high demand that impact intervention implementation.
    - Document staff roles, including whether additional staff are dedicated to the intervention.

- **Socioeconomic and cultural influences**
    - Note any visible socioeconomic and cultural factors among staff or other participants that influence how the intervention is received or delivered.

- **Broader community engagement**
    - Observe any interaction with community members or external organizations that support or hinder intervention activities.

Fidelity and adaptation of intervention

- **Adherence to protocols**
    - Observe whether staff are implementing and following the intervention as it was designed and planned.
    - Document any variations from the established protocols and seek to understand if these deviations are due to necessity, convenience, or improvisation through discussion with staff members

- **Adaptations in real-time**
    - Note any adjustments staff make to fit the intervention within existing routines or due to facility limitations.
    - Document the reasons for adaptations, noting if they are one-time adjustments or part of an emerging pattern.

- **Pause points and adjustments**
    - Document the exchanges during the pause-points designed to assess and refine the intervention
    - Document meetings, discussions, and logistics for launching and maintaining the intervention; the frequency, consistency

Responses of participants and key actors

- **Participants' physical health, mental health & well-being, performance and quality of care they provide**
    - Pay attention to these key areas of our study to notice any practices, or to discuss when making small talk during participant observation

- **Engagement of staff and participants**
    - Observe how actively staff participate in delivering the intervention, including their attentiveness and motivation.
    - Document participants' reactions—positive or negative—including any verbal or non-verbal cues that indicate their comfort or discomfort.

- **Acceptance and satisfaction**
    - Note whether staff and participants express satisfaction or dissatisfaction with the intervention's procedures and objectives.
    - Listen for and document feedback on aspects of the intervention that staff or participants find helpful, challenging, or problematic.

- **Communication and interactions**
    - Observe communication between staff and participants, noting if instructions, objectives, and benefits are clearly communicated and understood.

Unexpected outcomes and consequences

- **Unanticipated positive effects**
    - Observe any unexpected positive impacts on the facility, staff morale, participant engagement, or surrounding community.
    - Note if any component of the intervention appears to spark increased collaboration, improved efficiency, or better patient outcomes.

- **Unintended negative impacts or trade-offs**
    - Document any adverse effects such as increased staff stress, conflicts, disruption to other facility routines, or participant dissatisfaction.
    - Observe if any group (e.g., staff or participants) is disproportionately affected in unintended ways, and note their reactions.

Relations between staff, patients and companions in the hospital (if time permits)

- Reception of patients (how are they received? Are they informed and oriented/directed? By whom?)
- Nature of interactions/relationships [do patients and staff talk to each other? language used, use of an interpreter, topics of discussion, influence of the social level of patients in the process of their care (poor/rich, urban/rural, ethnicity, etc.)]
- Empathy, respect and explanations given to the patient and her family – note time given and patience expressed on the part of all
- Freedom given to the woman in relation to her preferences for giving birth (positions, right to walk, support and role of companions)
- Relations with accompanying persons (staff attitudes towards accompanying persons, information given and requests)

- Care and advice for the mother and the newborn (see what types of care and advice are given to the woman after childbirth, for the maintenance of the baby and for herself)
- Care and follow-up of the patient and newborn during and after childbirth (is the patient followed up/accompanied?)

Additional notes and reflections

- **Critical incidents**: Record any notable events that illustrate a specific challenge or success in the intervention process.

- **Personal reflections**: Capture your reflections, noting potential biases and assumptions that may influence your observations.

4c: Guide for focus group discussion with facility staff

FGD identification number:	
Facility:	
Date:	
Description of place where discussion took place:	
Interviewer/moderator's name:	
Note taker's/recorder's name:	
Start/finish time:	
Length of time taken:	
Participants: Socio-demographics	

	Pseudonym / ID number	Age	Occupation / role	Notes
1				
2				
3				
4				
5				
6				
7				
8				

Record of observations during FGD:

Include: time stamps for notable statements or occurrences during discussion; observations on dynamics amongst participants and facilitators; interruptions or arrivals/departures of participants; body language of participants and facilitator; notes to questions that worked well or didn't seem to work well, etc.

Introduction

- Share ground rules and confidentiality agreements, emphasizing respectful discussion and open sharing

1. Personal heat experience (potential icebreaker and warm-up questions)

- **Ideal temperature range**
  - What is the ideal temperature range for you to work and live comfortably day-to-day?
  - *Follow-up*: Do you think this range changes depending on the activity, like recreation versus work? If so, why?
- **Impact of extreme heat on daily life**
  - Does extreme heat affect your daily life? If yes, how does it affect different areas like home, commuting, and work?
  - *[Prompts: Encourage group members to describe any physical, mental, or logistical challenges they associate with heat in each setting.]*

2. Co-design process of the intervention

- **Involvement in co-design**
  - How many of you were involved in the co-design of the heat intervention? What was your role, if any, in the process?
  - *[Follow-up prompt: Ask participants who were involved to describe how the intervention development was conducted.]*
- **Inclusivity and decision-making**
  - Do you feel the right people were involved in the design process? Were everyone's voices heard?
  - *[Prompt: Discuss whether participants feel that certain voices were more influential or if additional perspectives should have been included]*
- **Factors considered**
  - What factors do you feel were important in creating this intervention, and do you think they were considered in the planning?

3. Implementation of the intervention

- **Consistency in implementation**
  - What do you think about how the interventions have been implemented? Do you think it's happened/happening as intended?
    In which ways?
- **Factors affecting implementation**:
  - What do you believe helps or hinders the intervention from being fully implemented?

4. Experience with intervention components

- **Discussing each component**
    - Let's go through each part of the intervention together. What are your thoughts on each component?
    - *[Prompts: For each, ask about team acceptance, feasibility, integration into daily practices, and effectiveness in reducing heat strain.]*

5. Impact of the intervention

- **Overall impact**
    - Do you think the intervention has made a difference in your work and well-being?
    - *[Prompt specific areas: Discuss changes in work performance, physical/mental well-being, and service delivery.]*
- **Patient perceptions**
    - Do any of you sense that patients have noticed a difference since the intervention started? In what ways?

6. Improvement suggestions

- **Refinements and future changes**
    - What changes or improvements would you suggest for this intervention if it were to continue?
- **Other potential interventions**
    - Are there other interventions you feel might be useful in this setting?

7. Contextual factors

- **Supporting and hindering contextual factors**
    - What factors within the facility do you think have supported the intervention?
    - What factors, if any, outside of the facility do you think have supported the intervention?
    - Are there any factors that you feel hindered it?

8. Sustainability of the intervention

- **Sustained impact**
    - How likely do you think it is that the changes made will still be in place 3 months or 3 years from now?
- **Program continuity**
    - Do you think the program is sustainable and should be continued?

9. Additional reflections and closing

- **Unintended consequences**
  - Have you witnessed or experienced any unintended consequences of these interventions? If so, which ones?
- **Comparisons to other interventions**
  - Are there any other programs or interventions you've been involved in at work that you feel have made a positive impact? Why do you think those were successful?
- **Reflections on heat stress awareness**
  - What do you think about the attention given to heat stress in your facility now? Do you feel it's beneficial to raise awareness on this topic?
- **Recommendations**
  - What are your recommendations to improve the program your facility has undergone to improve heat adaptation?

Final thoughts

- Is there anything else you'd like to share about your experience with the interventions or with managing heat?

15.5 Appendix 5: Cost tools

The cost tool consists of several spreadsheets to be completed.

Tool for estimating costs of the adaptation-mitigation interventions in South Africa and Zimbabwe as part of the HIGH-Horizons project

This tool has been developed to estimate total costs of developing and implementing the climate change adaptation and mitigation interventions. Outputs are:
(1) **Total resources** used for setting up and running the interventions
(2) Estimates of the costs of particular **activities,** such as community mobilisation, training and monitoring & evaluation

Add rows as required for based on the number of interventions evaluated

	List of intervention
1	Heat Champions
2	
3	
4	
5	
6	
7	
8	

Instructions:
1. Please enter the data in each tab.
2. Data in each tab should be entered for the single interventions which should be specified
3. Please provide a description of each intervention in the "intervention description" tab. Include any activities involved for each internvention
4. Each Excel file should be used for single facilities that are included in the costing - the facility should be described in the tab "Facility description"

Methodology:

Capital versus recurrent costs
Capital costs are items that last longer than one year, such as equipment.
Recurrent costs are those items that are used up during a year and usually purchased regularly - these include both medical and non-medical consumables.
To be able to combine capital and recurrent costs in a useful manner, all costs must be presented on an annual basis.
Capital costs are annualised by spreading their value over their expected lifetime. Future values are discounted to account for the fact that resources are tied up in the items.

Overhead costs
Overhead costs are resources used for the overall running of the facility, which do not vary much with the number of patients . These include administration, laundry services, electricity, etc. and can be found in the utilities tab

Description of the interventions

Have one table for each intervention- please feel free to add lines for the different activities and ingridients if needed.

Intervention name	Description	Adaptation/mitigation	Location of implementation	Start date	Time period	Activities	Ingredients
Heat Champions							

Intervention name	Description	Adaptation/mitigation	Location of implementation	Start date	Time period	Activities	Ingredients
Cooling vests							

Intervention name	Description	Adaptation/mitigation	Location of implementation	Start date	Time period	Activities	Ingredients
Air conditioning							

Description of the health facility

This refers to the facility where the intervention is being implemented

Name of facility:	

Facility ownership:	**Yes/No**	**Specify ownership**
Government		
Church		
NGO		
Private for profit		

Size of catchment population:	
Describe catchment population	

Physical facilities:	**Yes/No**	**Number**	**Shared with other services? Yes/No**
Registration office			
Outpatient department			
Waiting area			
Triage room			
Consulting room			
ORS corner			
Operating theatre			
Pharmacy			
Laboratory			
Accountancy office			
Storage room			
Sterilization room			
Administration office			
Seminar room			
Ward			

Number of hospital beds:	

Parameters assumptions

Add rows as required by analysis

Parameter name	Value	Source
Year of evaluation:		
No. of months evaluated:		
No. of days in a month:		
No. of days in a year:		
Exchange rate to US$		
Discount rate		

Capital costs

List all capital costs for each intervention

Intervention name									
Capital cost name	Quantity	Price for new (currency)	Price for new (US$)	Life expectancy (years)	Annualization factor	Annual cost (currency)	Annual cost (US$)	Annual cost allocated to intervention (currency)	Annual cost allocated to intervention (US$)

Cost of supplies

List all capital costs for each intervention

Intervention name					
Supply name	Quantity	Unit cost (currency)	Unit cost (US$)	Annual cost (currency)	Annual cost (US$)

List all capital costs for each intervention

Intervention name					
Supply name	Quantity	Unit cost (currency)	Unit cost (US$)	Annual cost (currency)	Annual cost (US$)

Staff salaries

List names of all staff positions for each intervention

Intervention name								
Staff involved In the intervention	No.	Monthly salary (currency)	Monthly benefits (currency)	Total (currency	Total (US$)	Percent of time working	Costs allocated (currency)	Costs allocated to U5 pneumonia (US$)

Intervention name								
Staff involved In the intervention	No.	Monthly salary (currency)	Monthly benefits (currency)	Total (currency	Total (US$)	Percent of time working	Costs allocated (currency)	Costs allocated to U5 pneumonia (US$)

Annual utility and insurance costs

Intervention name	

Please add any rows as required for the different interventions

	(currency)	US$
Annual electricity expenses		
Annual water expenses		
Annual telephone expenses		
Annual insurance expenses		
TOTAL		

Outcomes

Please provide a description of the outcomes measured for each intervention evaluated

Intervention name	

Outcome measure description	

Outputs description	

Number implemented	